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Recall length and measurement error in agricultural surveys

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ABSTRACT

This paper assesses the relationship between the length of recall and nonrandom error in agricultural survey data. Using data from LSMS-ISA surveys in Malawi and Tanzania, we show that key input and output variables are systematically related to the length of the recall period, indicating the presence of nonrandom measurement error. With longer recall periods, farmers report higher quantities of harvest, labor and fertilizer inputs. Farmers list fewer plots as the recall period increases. We argue that it is plausible that farmers over-estimate plot-level outcomes – harvest, labor and fertilizer inputs – while it is also plausible that they forget some of their more marginal plots as their memory decays due to longer recall periods. We find evidence of measurement error related to the length of recall also in common measures of agricultural productivity. The size of the recall effect typically varies between 2 and 5 percent per additional month of recall length, making its impact on the reliability of key agricultural indicators, including SDG indicator 2.3, economically significant. With data reliability affecting policy effectiveness, improving agricultural survey data quality remains an important concern, especially in light of the ambitious 2030 agenda reflected in the Sustainable Development Goals. Mainstreaming objective measures where possible and reducing the risk of recall error through shorter recall periods appear to be promising avenues to improve the quality of key variables in agricultural surveys.

1. Introduction

For many developing countries, boosting agricultural productivity and incomes of small food producers are key policy objectives in the effort to alleviate poverty and improve food security. High-quality, reliable farm output and input data are indispensable both for measuring progress towards the Sustainable Development Goals (SDG)[2] and for effective policy and program design and analysis.

Household and farm surveys are the most important source of agricultural data, rivalled in some countries by administrative data. In all measurement, survey-based or otherwise, there is some amount of measurement error. In surveys, the necessary reliance on imperfect respondent recall is a common source of error. Measurement in agricultural surveys in low-income countries is particularly difficult because agricultural operations are complex and seasonally variable, respondents are often illiterate, familiarity with standard units of measurement is the exception rather than the norm, and a host of other factors. When data are of poor quality, and particularly when they are affected by systematic biases, policies based on that information may turn out to be less effective, if not misguided.

Methodological research shows that survey design matters and that conscious efforts to adopt improved design and survey administration choices can contribute significantly to minimizing measurement error, including recall error (Kasprzyk, 2005; Carletto et al., 2015; Zezza et al., 2016; Gaddis et al., forthcoming, 2020; and Kilic et al., 2018). This paper contributes to this literature by focusing on the length of the recall period and how it affects the reliability of agricultural survey data. The length of the recall period is thought to be associated with recall decay, in which survey respondents forget details of events or forget events entirely, leading to reporting inaccuracies (Sudman and Bradburn, 1973). In agricultural surveys, recall length also often differs across domains of measurement because farming activities (land preparation, planting, harvest) take place at different times during the agricultural year. Depending on the timing of survey administration, respondents

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[2] Especially relevant are indicators 2.3.1 *Productivity of small-scale food producers* and 2.3.2 *Income of small-scale food producers.*

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may be asked to recall the details of events sometimes many months in the past, and the recall period may vary substantially within the same sample if the fieldwork is completed over several months. In countries with two or more cropping seasons agricultural activities can vary in extent and nature depending on the season, making it even harder for respondents to recall events correctly.

Survey designers and planners have some control over the length of the recall period by choosing the number and timing of field visits to farms. This choice, however, has clear cost implications. Visiting households or farms multiple times in order to shorten the length of the recall period is more expensive and logistically demanding than collecting information only once during the agricultural year. To evaluate the trade-off between quality and cost, survey designers require an empirically-informed understanding of the extent to which data quality varies with differences in the recall period, among other data quality considerations. Policy makers and other data users benefit from an appreciation of the limitations and biases in existing data related to recall length.

Previous work by Beegle et al. (2012a) found no consistent evidence of significant recall length effects in agricultural surveys, using data from the early 2000s. In contrast, in this paper, we find a significant effect of the recall length on key agricultural statistics. We take advantage of the recent availability of high-quality and very detailed agricultural survey data to overcome some of the data limitations of previous studies and assess and quantify the impact of the recall length on the quality of agricultural data. We use data from nationally representative surveys conducted by the National Statistical Offices of Tanzania and Malawi with support from the World Bank's Living Standards Measurement Study – Integrated Surveys on Agriculture (LSMS-ISA) program, and match survey interview dates with information on the timing of different agricultural activities, to construct household- and variable-specific measures of recall length. Exploiting the variability in recall length, we find that longer recall periods are associated with overreporting of plot-level outcomes – production quantity, labor input, and fertilizer input – consistently across the three datasets in the analysis. At the same time, respondents appear to forget listing plots as the recall period increases. Our results point to the need to devise and promote the adoption of survey design choices that minimize this source of measurement error.

The remainder of the paper is structured as follows: Section 2 reviews the literature on error in the measurement of agricultural outcomes, especially that related to recall length. Section 3 describes the data used, the variables generated, and the empirical strategy this analysis employs. Section 4 presents the main results. Section 5 discusses policy implications and offers recommendations and conclusions.

2. Background and related literature

This paper contributes to a large body of literature exploring the effect of respondent recall on measurement accuracy in survey data. Systematic reviews of this literature are provided in Biemer et al. (2011), Bound et al. (2001), Groves and Lyberg (2010), and Meyer et al. (2015), among others. The role of recall error in survey data has been explored in a range of topics, including consumption expenditure and food intake (e.g. Beegle et al., 2012b; Troubat and Grünberger, 2017; Backiny-Yetna et al., 2017; Brzozowski et al., 2017; Engle-Stone et al., 2017; D'Alessio, 2017; Schündeln, 2018; Zezza et al., 2017), health outcomes (Das et al., 2012), household enterprises (De Mel et al., 2009; Liedholm, 1991) and income measurement (Moore et al., 2000).

One strand of the literature focuses on the cognitive processes underlying survey response and its relationship with measurement error. Sudman and Bradburn (1973) distinguish between *recall decay* and *telescoping*. Telescoping refers to inaccurately remembering *timing* of events, leading to an event of interest incorrectly being moved into the reference period, which, in turn, may lead to over-reporting. In recall decay, respondents forget details of events or forget events entirely. The length of the recall period is arguably the most important driver of recall decay. As the recall length increases, recall decay occurs, and respondents no longer carry the requested information in memory, they turn to *reconstruction* strategies (Moore et al., 2000). Common reconstruction strategies include basing the response on typical behavior or average circumstances, which may lead to reporting error. Whether such error is nonrandom depends on the type of reconstruction strategy and may be related to respondent characteristics, such as cognitive ability and education. Reconstruction is also associated with *rounding* of quantities, which is common in all domains of subjective reporting (Roberts and Brewer, 2001) and for agriculture has been documented for instance with respect to land area measurement by Carletto et al. (2011). Finally, the *salience* of events can counteract recall decay as respondents are able to recall salient events more easily, though some research has found salience being associated with over-estimation (see Bound et al., 2001, for a discussion).

There is some evidence of the effect of recall error on each of the outcomes of interest of this study – land input, crop production, labor and agricultural input use in low-income settings. Beegle et al. (2012a) explore the effect of recall length on the reliability of both agricultural input and output data, while others have focused on error affecting particular aspects of measurement of the agricultural production process (e.g. Arthi et al., 2018; Gaddis et al., forthcoming, 2020; Kilic et al., 2018; Zezza et al., 2016).

Regarding the measurement of **agricultural land**, the biases in farmer-reported land area estimates have attracted the most attention over the past 15 years (e.g. Carletto et al., 2011; De Groote and Traoré, 2005; Dillon et al., 2019; Goldstein and Udry, 1999; Keita and Carfagna, 2009; Kilic et al., 2017; Schøning, 2005). None of these studies, however, focus on the role of recall length in affecting the error in farmer-reported land area. Two recent studies, Arthi et al. (2018) and Gaddis et al. (2020, forthcoming), find that respondents tend to undercount the number of plots under cultivation when using end-of-season recall to elicit the information.

With respect to **crop production** measurement, Beegle et al. (2012a) find no consistent evidence that the length of the recall period affects farmers' harvest estimates in data from Kenya, Malawi, and Rwanda from the early 2000s. In contrast, Deininger et al. (2012) focusing on a wide range of crops in Uganda find that harvest quantities based on end-of-season recall diverge significantly from those recorded in continuously administered harvest diaries, which are deemed more reliable. In most cases, end-of-season recall is associated with under-reporting, especially for extended-harvest crops such as cassava or banana. Cash crop production, in contrast, is significantly over-reported. Similarly, Kilic et al. (2018) compare weekly harvest diaries to a 12-month recall period and a 6-month recall period each. All methods are benchmarked against crop-cutting estimates, considered the 'gold standard' for estimating crop production. Their results show significant under-reporting in the recall data, especially with the 12-month reference period, relative to the diary and crop cutting methods.

A different strand of the literature found that reported harvest quantity is inversely related to land area: respondents tend to over-report maize harvest on small plots and under-report it on large plots (Desiere and Jolliffe, 2018; Bevis and Barrett, 2019; Gourlay et al., 2019). Relatedly, Abay et al. (2019) looking at measurement error in both production and plot size show how, when non-classical measurement error is correlated across variables on either side of a regression, correcting for just one source of error can actually aggravate bias. Different variables in a survey sharing underlying error-generating mechanisms related to the length of the recall period, which is the issue we investigate, is a potential major source of correlated measurement error analysts should be aware of.

Beegle et al. (2012a) assess the effect of recall length on farmer reported **labor inputs**, finding no consistent evidence of recall decay affecting reported quantities. Arthi et al. (2018) and Gaddis et al. (2020, forthcoming), on the other hand, find significant measurement error in

recall-based farm labor estimates. The two studies compare end-of-season recall with weekly work diaries, which are considered more reliable, among farmers in Tanzania and Ghana. They find that end-of-season recall leads to over-reporting of labor use at the individual-by-plot level ('recall bias') relative to diary-keeping, but at the same time to undercounting of plots cultivated and of individuals who worked on them ('listing bias'). In contrast, Gollin (2019) argues that diary methods may lead to under-counting of farm labor given the eclectic set of tasks that 'farm labor' comprises. Seymour et al. (2017) find significant differences in the recording of time spent working between recall and diary methods in Uganda and Bangladesh.

Empirical evidence on recall error (and measurement error in general) in **agricultural inputs** data (fertilizer, agro-chemicals, seeds, and others) is scant. Beegle et al. (2012a) find little systematic evidence for recall bias in farmer self-reported fertilizer usage in Malawi and Kenya, though results vary somewhat depending on respondent characteristics. Gollin (2019) argues that farmers likely recall quantities (and prices) of purchased inputs mostly accurately but also points to evidence (e.g. in Ashour et al., 2017; Bold et al., 2017) suggesting that fertilizer and agro-chemical counterfeiting and adulteration may leave farmers unsure about the quality of product they apply to their land.

3. Data and empirical strategy

3.1. Data

The study uses three datasets from nationally representative household surveys in Malawi and Tanzania: the Tanzania National Panel Survey (TNPS) 2012/13, the Fourth Malawi Integrated Household Survey 2016/17 (IHS4) and the Malawi Integrated Household Panel Survey 2016/17 (IHPS).

Tanzania NPS 2012/13 is the third wave of the Tanzania National Panel Survey and was collected between October 2012 and November 2013. The NPS sample used a multi-stage clustered sampling design covering a total of 5,015 urban and rural households. With households selected to be interviewed over the course of 14 months to account for seasonality in consumption, the data have variation in the recall period, which can be exploited to assess the impact of recall decay on data quality. The reference agricultural season is the long rainy season of 2012, running approximately from February through August, which all farms in the sample reported on. At the time of the interviews, all farms had finished their maize harvest for the reference agricultural season.

The Malawi IHS4 2016/17 is a cross-sectional sample of 12,480 households surveyed in 780 enumeration areas. Households were visited once throughout the 12 months of fieldwork between April 2016 and April 2017. The IHS4 data was collected in the same way as the NPS 2012/13 and used for analysis following the same rationale. For our analysis, we use the 2015/16 rainy season, which ran approximately from November 2015 to May 2016,[3] as the reference agricultural season. To create a comparable sample, we kept all households that had finished the 2015/16 rainy season harvest at the time of the interview in full and dropped all others.[4]

The Malawi IHPS 2016/17 sample includes 1,989 panel households which were interviewed twice between April 2016 and April 2017, in one post-planting and one post-harvest visit relative to the 2015/16 rainy season. Given this two-visit setup, the recall length is shorter, especially for planting-related activities. Including this dataset first allows us to assess if recall effects vanish with somewhat shorter recall periods. The fact that data was collected in two rather than one visits may have an effect on reporting beyond simple differences in recall length. Unfortunately, we cannot distinguish between the recall length effect and the effect of fielding two visits with the data at hand.

All three surveys are part of the Living Standard Measurement Study – Integrated Survey on Agriculture (LSMS-ISA) and, as such, the datasets contain an integrated household and agricultural component. The household survey component collects detailed socioeconomic information, including household-level data on consumption, income, assets and housing, and individual-level data on demographics, education, and health. The agricultural component collects detailed information on agricultural inputs used and outputs produced, as well as output disposition, at the plot-level. The integrated survey further allows breaking down which household members are involved in agricultural production, who owns the means of production, and who is responsible for managing the household farm's plots of land.

As ancillary data to validate our findings on farmer-reported crop production, we use the second round of the Methodological Experiment on Measuring Maize Productivity, Soil Fertility and Variety (MAPS) survey. The MAPS data, which Gourlay et al. (2019) discuss in detail, were collected in two rounds in 2015 and 2016 in Uganda, containing 900 maize-growing households in round 1 and 489 in round 2. The MAPS survey employed objective and subjective survey methods to collect information on maize production (crop cutting), area, soil fertility, and maize variety identification.

3.2. Outcomes of interest

The analysis focuses on some of the main agricultural outcomes of policy interest, namely land input, crop production, labor and input use. The main independent variable is the approximate length of the recall period between the interview and the activity relating to the outcome of interest (Table 1).

Land input reporting is assessed in two ways. First, as the total count of the households' agricultural plots (or parcels) which respondents list at the beginning of the agricultural questionnaire. This information sheds light on whether a longer recall period leads households to forget listing some plots (Gaddis et al., 2020, forthcoming). The median (mean) number of plots listed per household is 2 in Tanzania NPS 2012/13 (mean: 2.2 plots) and Malawi IHPS 2016/17 (mean: 2.1 plots) and 1 in Malawi IHS4 (mean: 1.6 plots; Table A.1).

Second, as the difference between GPS-measured plot area and farmer self-reported plot area. LSMS-ISA surveys implement an objective, GPS-based area measurement, while retaining farmer self-reported plot area alongside. Comparing these two measures allows assessing whether recall decay exacerbates the error in self-reported land area. This variable is constructed as the absolute difference between GPS-based and self-reported plot area variable.

Plot sizes, measured by GPS, differ substantially across the three datasets. In Tanzania NPS 2012/13, the average plot size of 1.23 ha (median: 0.49 ha) is much larger than in Malawi at 0.37 ha in the IHS4 (median: 0.30 ha) and 0.34 ha in the IHPS dataset (median: 0.26 ha; Table A.1). The inverse relationship between measured plot size and error in self-reported plot size discussed in the literature is also common to all three datasets used in this analysis.[5]

[3] Earlier or later harvests are possible depending on the type of crop, seasonal and regional variations in rainfall and other location- and time-specific agronomic and climatic conditions.

[4] Data collection began in April 2016 before the end of the season, so not all households in the IHS4 survey had finished the harvest in full. Some households had not started the 2015/16 harvest and instead reported on the previous 2014/15 rainy season harvest. Another group of households had started but not completed the 2015/16 maize harvest in full. These two groups of households were dropped. In addition, some households stated having harvested in the 2015/16 season, but harvest dates revealed respondents were referring to the previous season, and vice versa. These households were dropped as well.

[5] The inverse relationship holds also for the other outcomes of interest: maize yield (output per hectare); labor intensity (labor input per hectare); fertilizer intensity (fertilizer input per hectare). The farmer-reported measurements of these variables are decreasing in plot size.

Table 1
Outcome and main explanatory variables of interest.

	Dependent variable (Y_i)		Unit of measurement	Level of analysis	Main independent variable (T_i)
Output/ Production	Maize quantity produced		kg	Plot-level	Recall length (months) to harvest end
	Harvest disposition	Maize quantity sold Maize in storage		Household/farm-level	
Inputs – Land	Agricultural plots and parcels listed and cultivated by the household		number	Household/farm-level	Recall length (months) to planting of plot*
	Difference between self-reported and GPS-measured plot land area		hectares	Plot-level	
Inputs – Labor	Total labor inputs by type, household and external	Total labor	person-days	Plot-level	Recall length (months) to planting of plot*
		Household member labor			
		Use of hired labor	binary		
	Labor by activity	Plot preparation and planting	person-days	Plot-level	Recall length (months) to planting of plot*
		Plot maintenance			
		Harvesting and post-harvest activities			Recall length (months) to harvest end
Inputs – other	Organic fertilizer use		binary	Plot-level	Recall length (months) to planting of plot*
	Inorganic fertilizer use				
	Inorganic fertilizer application quantity		kg		Recall length (months) to first fertilizer application*

* In the Tanzania NPS 2012/13 survey, these variables are not available at the plot-level. Regional sowing season timing information is used instead.

Crop production is assessed, first, through the quantity of maize harvested per plot. We choose maize since it is the most important staple crop in Malawi and Tanzania, though our findings on maize do not necessarily translate to other crops (see results in Deininger et al., 2012). In addition, we analyze the amount of maize harvest stored and the amount sold at the time of the interview.

Harvest quantities are recorded separately for each crop and plot, and all surveys allowed for reporting in local non-standard units, converted into kilograms using correspondence tables specific to each survey. Harvest is reported by respondents at the plot level, and we do not aggregate it nor normalize it by unit of land area, so that the variable we analyze best reflects the cognitive process of the respondent who reported it. Stored and sold harvest is reported at the household level. The crop production variables are summarized for each dataset separately in Table A.1. The median maize harvest quantity per plot is 200 kg in Tanzania NPS 2012/13 (mean: 303.7 kg), 213.4 in Malawi IHS4 (mean: 320.5 kg), and 180 kg in Malawi IHPS 2016/17 (mean: 286.2 kg). While harvest per plot is similar, maize yield (harvest per area) is higher in Malawi IHS4 and IHPS 2016/17 than in Tanzania NPS 2012/13, where plots are larger on average (Table A.1).

The three main surveys used in this analysis rely on farmer self-reporting of the quantity of maize harvested. This means that there is no approximately objective measure of maize harvest to benchmark the results. We therefore make use of ancillary data from the second round of the Methodological Experiment on Measuring Maize Productivity, Soil Fertility and Variety (MAPS). These data contain full crop-cuts (considered benchmark estimates) for maize of 211 entire plots. The data set has two main limitations for our analysis: the data collection was so swift that the recall length is short with little variation between 0 and 4 months, and, at 211 plots, the sample size is small. The full crop cuts are used to quantify the error in farmer-reported maize harvest and relate it to the length of the recall period. On those plots, the mean farmer-reported harvest is 159 kg (median: 70 kg), while the mean crop cut estimate is 113.5 kg (median: 53.45; Table A.3)

For farm labor inputs, the headline outcome is total person-days of labor used per plot during the reference agricultural season. Family labor makes up the lion's share of total labor input in all three datasets, accounting for between 91 percent in Tanzania NPS 2012/13, 92 percent in Malawi IHPS 2016/17, and 96 percent in IHS4 (cross section). Total labor input is somewhat higher in Tanzania at 70 person-days than in Malawi with 64 and 62 person-days per plot in the cross section and the panel, respectively. However, given larger plot size in Tanzania, labor intensity (labor per area) is lower there than in the two Malawi samples (Table A.1). 26 percent of plots in Tanzania NPS 2012/13 use hired labor, compared to 13 percent in Malawi IHS4 and 29 percent in IHPS 2016/17 (Table A.1).

For an analysis of recall length, the total person-days variable has some drawbacks. As it is constructed by summing various work activities (planting, fertilizing, harvest, etc.), which all take place at different times during the agricultural season, there is no one single recall period for total labor person-days. To account for this, we assess whether the effect of recall length differs by type of activity (planting, fertilizing, harvest, etc.) using appropriately defined recall length variables for each activity (Table 1, Table A.2). Of the activities, most person-days are used in planting and maintenance activities, with fewer person-days used in harvest (Table A.1). We also assess whether the recall effect differs by type of worker (household, hired). Labor inputs are summed to the plot-level, rather than assessed separately for each worker on each plot, because one respondent reports all labor inputs for a given plot, so that this measure likely best captures the recall effect.

Finally, input use is assessed through incidence of organic fertilizer application per plot and incidence and quantity of inorganic fertilizer application per plot. Fertilizer use is considerably more common in Malawi than in Tanzania, for which Malawi's inorganic fertilizer subsidy program is likely responsible. Organic fertilizer is applied on 19 and 20 percent of plots in Malawi IHS4 and panel, respectively, and on 10 percent of plots in Tanzania NPS 2012/13. The difference is larger for inorganic fertilizer, which 53 and 61 percent of plots receive in Malawi IHS4 and IHPS 2016/17, respectively, compared to 9 percent in Tanzania[6]. Among those plots to which inorganic fertilizer is applied, the applied quantities are similar across all three datasets at 66 kg per plot in Tanzania NPS 2012/13, 62 kg per plot in Malawi IHS4 cross-section, and 58 kg per plot in IHPS 2016/17. Application intensity (inorganic fertilizer applied per hectare) is again lower in Tanzania than in Malawi (Table A.1).

We apply the double Median Absolute Deviation method for outlier detection and correction to all constructed continuous variables in this

[6] Fertilizer use is on the increase in Tanzania as well. In the most recent year for which NPS data are available (2014/15) inorganic fertilizer is used on 11 percent of the plots, and the quantity applied per plot is 73 kg per plot.

analysis, that is, harvest quantity, storage and sales quantity, total, family, and hired labor person-days, inorganic fertilizer quantity applied (Leys et al., 2013).

3.3. Determining recall length and identifying its effect

The length of the recall period cannot be determined exactly in the survey data because we do not observe directly when farmers engage in the activities of interest (planting, fertilizing, working, harvesting, etc.), and because farmers engage in these activities over extended time periods, so that strictly speaking no single recall length exists. We therefore proxy the length of the recall period for each outcome of interest using information about the timing of the activities sourced from farmer-reported dates and the cropping calendar.

For harvest (and storage and sales) quantity, we use as recall length the distance in months between the interview date and the end of the harvest period. In all three surveys, farmers report harvest dates and this information is used to construct the recall length variable. Harvest dates are likely recorded with error, being subject to memory decay over time in the same way as the outcome variables. To address this concern, we use the mode of harvest end dates at the cluster (enumeration area) level, rather than each household-plot's individual maize harvest date, provided that there are at least 10 observations per cluster. Where there are fewer than 10 observations per cluster, the variable is instead based on the next higher level of geographical aggregation (ward, district).[7] The interview date varies at the household level, so that the recall length variable for harvest for household i is

$$T_i = Recall\ \ length_i^{Harvest} = \left(Date_i^{Interview} - mode_z^{harvest\ \ end\ \ dates}\right) \tag{1}$$

with $mode_z^{harvest\ \ end\ \ dates}$ being the mode of the plot-level maize harvest dates for each cluster z. Table A.4 in the appendix presents an overview of the differences between the raw and corrected recall length variables for the three datasets.

We use a similar approach in determining the recall length for the other key variables of interest. In the analysis of land inputs (plots listed), the recall length is set with respect to the beginning of the planting season. In Malawi IHS4 and IHPS 2016/17, farmers report the month in which they began planting each of their plots. This information is used to capture the variation in season start. The recall variable is constructed following the same procedure[8] as for recall to harvest end, such that

$$T_i = Recall\ \ length_i^{Planting} = \left(Date_i^{Interview} - mode_z^{planting\ \ plot\ \ dates}\right) \tag{2}$$

with $mode_z^{plantingplotdates}$ being the mode of the plot-level dates of the planting start for each cluster z.

The Tanzania NPS 2012/13 does not record individual planting dates. Instead, we rely on regionally disaggregated maize sowing time information from Arce and Caballero (2015) and the FAO Crop Calendar. The recall length to planting start variable for household i in region k is therefore:

$$T_{ik} = Recall\ \ length_{ik}^{PlantingNPS} = \left(Date_i^{Interview} - Date_k^{Sowing}\right) \tag{3}$$

Proxying the length of the recall period for labor inputs is more challenging as labor in crop production is spread out across the entire cropping cycle and comprises a set of disparate activities, from plot preparation, to planting, to fertilizing and weeding, to harvest and post-harvest work. The activities differ in duration and salience and may be carried out in part by household members and in part by hired workers – which is why we assess not only total labor input but analyze each activity in detail. The timing of these activities is not always explicitly recorded in the surveys. Recall length variables therefore differ by labor activity and survey: for total labor input, we use recall length to planting from Eqs. (2) and (3), respectively. Since total labor is the sum of the various activities during the reference season, this recall variable is a rough approximation; for harvest and post-harvest labor, recall length to end of harvest, as per Eq. (1), is used; for planting and plot preparation, and weeding and fertilizing, we again use recall length to planting (Eqs. (2) and (3)).

Finally, constructing a sensible proxy for the recall length of fertilizer input has some of the same challenges as for labor, except that fertilizer input application covers shorter periods than the more extended task of working on the plot. In Malawi IHS4 and IHPS 2016/17, we use the recall length to planting (Eq. (2)) when analyzing input use in the full sample (that is, both plots on which fertilizer was used and plots on which it was not used). When assessing fertilizer quantity reporting among plots receiving fertilizer, we use the recall length to the first application of fertilizer on the plot. Farmers report the time of fertilizer application in weeks after the completion of planting. For each plot j of household i, we subtract the time between completion of planting and application of fertilizer from the recall length to planting:

$$T_{ij} = Recall\ \ length_{ij}^{fertilizer} = Recall\ \ length_i^{Planting} - time\ \ between\ \ planting\ \ and\ \ fertilizer\ \ application_{ij} \tag{4}$$

In Tanzania NPS 2012/13, we once again use recall length to planting (Eq. (3)), lacking information on fertilizer application timing.

All recall variables are coded in increments of one month as this is the unit in which most dates are recorded in the data.

Recall length to end of harvest ranges from 2 to 18 months in Tanzania NPS 2012/13 (mean: 9.2), 0 to 13 months in Malawi IHS4 (mean: 8.9),[9] and 3 to 12 months in Malawi IHPS 2016/17 (mean: 5.7; Table A.2, Fig. 1). Recall length to planting varies between 7 and 25 months in Tanzania NPS 2012/13 (mean: 14.7) and between 4 and 18 months in Malawi IHS4 2016/17 (mean: 13.4), though there are very few observations with a recall length shorter than 9 (Fig. 1); in Malawi IHPS 2016/17, recall length to planting varies between 4 and 9 (mean: 5.7; Table A.2). Because the IHPS has a post-planting visit, the average recall length to planting is much shorter than in the other surveys at 5.7 months relative to 13.4 (Malawi IHS4 2016/17) and 14.7 months (Tanzania NPS 2012/13). Recall length to first fertilizer application is minimally shorter than recall length to planting, suggesting most farmers start fertilizing quite soon after they finish planting (Fig. 1; Table A.2).

Since this analysis exploits variation in the timing of Tanzania NPS 2012/13 and Malawi IHS4 fieldwork, rather than explicitly randomizing by recall length to agricultural activities of interest, the data are not evenly spread across time and space (Fig. 1). Given the observational nature of our data, we need to ensure that any relationship we identify between the length of the recall period and the outcomes of interest is not due systematic differences between survey timing and household characteristics that may be correlated with the outcomes of interest. That would be the case if households interviewed later were located for

[7] This is the case in 11 percent of observations in Tanzania NPS 2012/13, 10 percent in Malawi IHS4, and 3.5 percent in Malawi IHPS.

[8] Mode of beginning of planting on plot at the cluster, as long as there are at least 10 observations per cluster. If fewer than 10 observations per cluster, the variable is based on the next higher level of geographical aggregation. This is the case in 5.2 percent of observations in Malawi IHS4 and 1.6 percent in IHPS 2016/17.

[9] There are three cases of a recall length to harvest end of zero months, which implies that in these cases farmers have finished their harvest within the month of being interviewed. This scenario can realistically occur in Malawi IHS4 because farmers are asked to identify the last rainy season completed, and the agricultural questionnaire is administered in reference to that season. The agricultural season spans a period of November to May and the three cases at hand are from households interviewed in April of 2016, so that it is very plausible that these farmers finished harvesting just before the interviews took place.

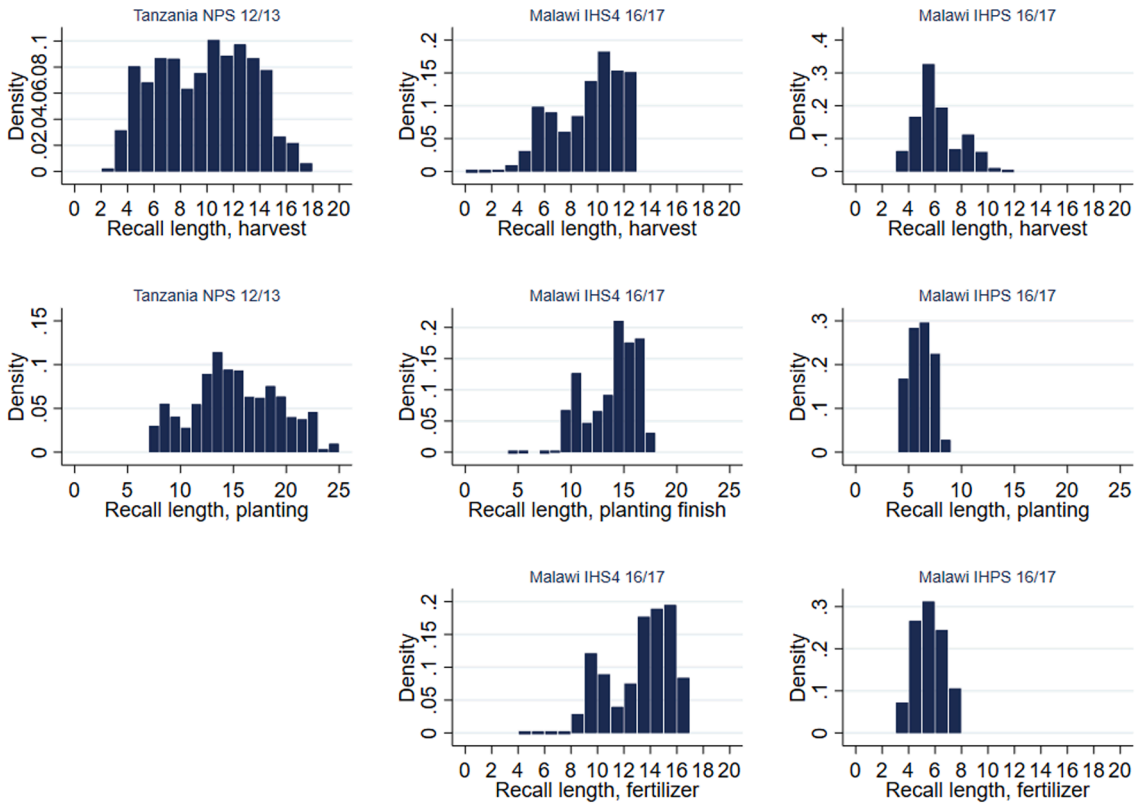

Fig. 1. Distribution of recall length variables.

Table 2
Control variables for regression analysis.

Group of control variables X_i	Variables Malawi IHS4 and IHPS4	Variables Tanzania NPS 2012/13
Respondent characteristics	Respondent gender, age, education (years), literacy.	Respondent age, gender, education (years).
Household characteristics	Drought shock, flood shock, income loss shock; agricultural asset index, household wealth index; household head literacy, gender, age; dependency ratio, household size.	Household size, production shocks, household consumption expenditure, household head gender.
Plot characteristics	Plot owned, cash crop grown, erosion control terraces, swampland, fertilizer use, plot size, any hired labor, family labor input, pre-harvest losses, intercropped.	Plot area; Main crop, irrigation, respondent is plot owner, respondent, or user; use of fertilizer; any hired labor; family labor input.
Regional controls	District dummies.	Region dummies.
Enumerator controls	Enumerator dummies.	Enumerator dummies.

instance in regions with larger farm sizes, more fertile land, or better access to markets for agricultural inputs and outputs. We would then expect these households to have truly higher level of agricultural output and input use, and any association between higher estimates of these outcomes and the recall period could not be ascribed to the latter.

Qualitative information on the modalities of survey rollout points to that not being the case. Staff intimately involved in the implementation of the three surveys[10] concur that the assignment of enumeration areas to different weeks of fieldwork is largely random. For the Tanzania NPS 2012/13, the sample is first split with half the sample randomly assigned to the first 6 month and the remaining in the second. This is purely random and intended to give a good geographic distribution at the halfway mark. While Tanzania's National Bureau of Statistics may use some discretion in the order of the specific enumeration of EAs, survey staff familiar with the operations confirm that lumping of interviews by area in time is unlikely to happen. In Malawi, the IHS4 2016/17 cross-section was randomly assigned into quarters and then months within those quarters. The timing of the IHPS 2016 panel interviews is less predictable because of the tracking of the split-off households, so in this case one may expect households that no longer reside in their baseline dwellings to have longer recall periods, though our results are robust to the inclusion or exclusion of these households (Section 4.5).

Quantitative descriptive information generally confirms that is the case. We split the survey samples in two groups, ordering households by timing of fieldwork implementation and creating one partition with those up to the 50th percentile, and one with those beyond that threshold. We then compare summary statistics of variables not based on respondent reporting which therefore have no links with the length of the recall period. These are variables related to household distance to roads, cities, and markets, as well as three climatic variables on average temperature, and short- and long-term annual average rainfall. We assess differences in these variables between the groups first at the EA-level, as this is the level at which fieldwork is rolled out. The results confirm that the two groups are not significantly different in these variables (Table A.5). We also report the results for the same exercise at the household level as this is the unit of observation for these variables

[10] Personal email communications from World Bank staff overseeing the technical assistance to the survey, available from the authors.

(Table A.6). In this case, there are some significant differences between the two samples. The differences do not however follow a systematic pattern: their direction is inconsistent, and different variables are significant in each data set.[11] We include these spatial variables as controls in the main regressions in the results section and, while some of them remain significant, they do not have any meaningful impact on the coefficient of the recall length variable. As an additional step, we test specifications with and without regional fixed effects and find the results to be robust to the inclusion of regional fixed effects (see discussion in Section 4.5).

Another factor potentially complicating the identification of the biases linked to the length of recall is enumerators' fatigue. Over the course of fieldwork, enumerators may tire altering their behavior and affecting the data in ways that would be correlated with recall length. For instance, tired enumerator may list fewer plots in the later months of fieldwork, and assign outputs and inputs to those fewer plots, thus creating a pattern in the data like the recall effect we find. While we do not rule out enumerators' fatigue, we argue that that is unlikely to fully account for the recall effect we find. At any point during fieldwork, how much and how long different enumerators have worked varies, as some of them do not work continuously, join fieldwork teams later, and take breaks, and as in general we do not expect all enumerators to tire in the same way. Enumerator controls should at least partially capture a potential fatigue effect, by absorbing some of the heterogeneity across enumerators. However, we do not find evidence of that when we test the sensitivity of our estimates to the inclusion of enumerators fixed effects (Section 4.5). At the same time, we cannot exclude that enumerator fatigue drives some of the effect we are finding, but argue that the policy implications remain the same whether recall decay alone or recall decay and enumerator fatigue are at play.

In the multivariate analysis presented in the following we introduce a set of control variables that fall into the following categories: regional and enumerator dummies, respondent characteristics (e.g. gender, age, education, whether respondent is plot manager), household characteristics (e.g. consumption aggregate or wealth index, shocks, household head gender, education, age), plot characteristics (e.g. crop loss, land area, irrigation; Table 2).

3.4. Estimating the recall effect

To assess the effect of the recall length on outcomes of interest we estimate the following equation:

$$Y_{ij} = \alpha + \beta_1 T_i + \delta \mathbb{X}_{ij} + \gamma + \mu + \varepsilon \tag{5}$$

with Y_i the outcome variable of plot ij (or household i depending on the level of analysis), T_i explanatory variable of interest (planting or harvest recall length in months, depending on the outcome of interest), $\mathbb{X}_{ij}$ a vector of control variables, γ an indicator for region, μ an enumerator fixed effect, and ε the error clustered at the EA level. Provided $\mathbb{X}_{ij}$ captures relevant confounding factors, the length of the recall period, T_i, should not be correlated with the outcomes variables if farmers accurately recall them. Therefore, if β_1, the coefficient on T_i, is significantly different from zero, this is evidence of systematic measurement error.

We estimate Eq. (5) with OLS when assessing the uncensored continuous outcome variables of interest, maize production, error in self-reporting, labor input in person-days. The number of plots listed and cultivated is a count variable, so we asses this outcome with OLS, Poisson, and negative binomial estimators. For the binary outcome variables use of hired labor, use of organic and of inorganic fertilizer, we make use of Probit. Finally, the quantity of inorganic fertilizer applied per plot is truncated at zero. On the considerable share of plots to which no fertilizer is applied, the fertilizer quantity variable takes the value zero, while it is continuous and positive for the share of plots to which fertilizer was applied. We deal with this case by running both a Tobit model on the truncated variable as well two separate regressions, a Probit model on the binary choice on inorganic fertilizer use and OLS on the continuous quantity of application conditional on use.

To verify that the linear baseline specification in Eq. (5) accurately captures the relationship between recall length and the outcomes of interest, we run a set of regressions with added polynomial terms (quadratic, cubic) of T_i (recall length). We compare the linear and non-linear models using a variant of the Bayesian Information Criterion (BIC') as a goodness-of-fit measure to guide which model is best suited.[12]

Recall error may be exacerbated by certain household, individual, or farm characteristics. We explore interaction effects between the recall length and plot size and a set of respondent characteristics (gender and education, whether the respondent is also the plot manager). The latter is relevant because, to an extent, survey designers have some control over who is the respondent.

4. Results

We find that the recall length has a significant impact on reported outcomes in all areas of interest of this analysis. Farmers report fewer cultivated plots of land with longer recall periods. In contrast, farmers report significantly higher quantities of plot-level variables, that is maize harvest, labor and fertilizer input, as the recall length increases.

4.1. Land

Households list significantly fewer plots (and parcels) as the recall period increases. Results from the baseline linear OLS specification with a full set of controls, shown in Fig. 2 and Fig. 3 (and Table A.7 in the Appendix), are consistent across the three datasets. The results from a negative binomial regression are in line with those from OLS. The magnitude of the effect varies by survey, however. It is small for Tanzania NPS 2012/13 at 0.03 fewer plots per month and Malawi IHS4 (cross section) at 0.05 plots per month, but larger in Malawi IHPS 2016/17 (panel) at 0.1 plots per month. The results from a quadratic specification confirm the overall negative relationship between recall length and plots listed (Table A.8).

We have no way to observe the 'true' number of plots household-farms in our data operate, so we cannot ultimately determine the direction of the reporting bias. However, it seems plausible that farmers forget some of their less important plots as recall decays over time, as was documented previously for example in Gaddis et al. (2020, forthcoming) and Arthi et al. (2018).[13]

For self-reported land area, we assess the effect of recall length on the error in self-reported plot area, which we proxy by the absolute difference between GPS measured area and self-reported area. We find a small positive association between the error and recall length in the Malawi

[11] For instance, in Tanzania the mean distance to a major road is lower for households interviewed later, but the mean distance to urban centers is higher; in the Malawi IHPS, opposite to Tanzania, the mean distance to major roads is higher for households interviewed later on.

[12] We display only the best fit models in Figs. 2, 4, 8, and 12.

[13] The analysis in this paper relies on a different setup than Arthi et al. (2018) and Gaddis et al. (2020, forthcoming), who use a methodological experiment in which farmers are randomly assigned to report on labor inputs either through end-of-season recall modules or in weekly visits during the agricultural season. In contrast, we rely on variation in recall length stemming from variation in interview timing. The findings in Arthi et al. (2018) and Gaddis et al. (2020, forthcoming) are therefore a combination of a recall effect (end-of-season modules imply longer periods of time to recall) and a data collection effect (weekly visits in which respondents are prompted to report in detail on their plots). Our findings isolate and underscore the former.

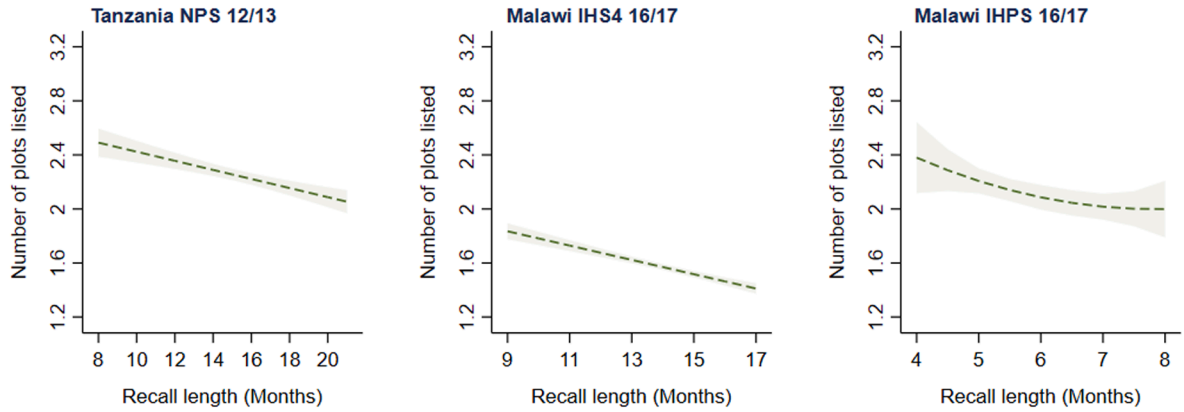

Fig. 2. Plots listed against recall length.

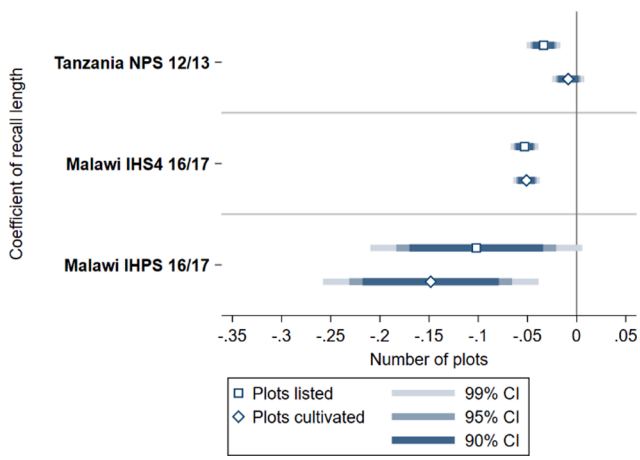

Fig. 3. Effect of recall length on plots listed and cultivated, point estimates.

IHS4 dataset, but no significant correlation in the other datasets (Table A.9). Overall, this suggests that respondents' difficulties in accurately estimating land area are not associated with memory decay over time. It is plausible that farmers at the time of the interview still operate many of the same plots as at the beginning of the season and therefore have a belief or estimate of plot area based on relatively current information.[14]

4.2. Crop production

The maize harvest quantity as reported by farmers increases with the length of the recall period. The results are consistent across all three datasets, with point estimates ranging from 4.9 kg (2 percent) in Tanzania NPS 2012/13, to 11.97 kg (4 percent) in Malawi IHS4, to 15.85 kg (8 percent) for each additional month of recall length in Malawi IHS4 panel (Fig. 5, Table A.10, Table A.11). Based on BIC', a quadratic specification yields a better fit of the relationship in Malawi IHS4, suggesting that here the recall length effect is strongest in the first 3 to 8 months, levelling off after that (Table A.12, Fig. 4). The results support the hypothesis that there is non-random measurement error in the farmer-reported quantity of maize harvested related to the length of the recall period.

Next, we assess the relationship between recall length and the quantity of maize sold and maize in storage (Table A.13 and Table A.14). The rationale for doing this is as follows: Absent measurement error, the quantity of maize harvested should not be correlated with the timing of the interview. In contrast, as the time between harvest and interview increases, the expectation is that households deplete their storage of harvested maize, while selling more of it. Thus, there should be a negative correlation between the time passed since harvest and maize in storage and a positive correlation with maize sold. Finding that this is not the case may point to some other underlying dynamic and cast doubt on the results. We thus test this hypothesis in the data. The data indeed reflect a negative relationship between time elapsed since harvest – that is, recall length – and the quantity of maize in storage. We also find a positive relationship between time elapsed since harvest and the quantity of maize farmers report they have sold.

Having found that reported harvest quantity increases with the recall period, the next question is whether this means that harvest quantity is actually over-reported as the recall period increases. The logic of recall decay would suggest this is the case: farmers likely remember the amount harvested if the harvest has only just finished. But absent written records, they need to rely on inference or reconstruction more often if the harvest is many months in the past. If this logic holds, interviews conducted just after the harvest, with a short recall length, produce the most reliable estimates. Whether or not this logic holds is predicated on respondents having accurate knowledge of how much they harvested in the first place. Existing evidence shows that other factors, such as land area or respondent characteristics, are correlated with harvest mis-reporting, so even in cases of short recall periods self-reported harvest quantities may well be inaccurate.

Resolving this question systematically would require comparing farmer-reported harvest quantities to more objective benchmark estimates, for example based on crop cutting, which however is not available in the three main datasets used for this analysis. As an ancillary analysis, we use the MAPSII dataset to benchmark the farmer self-reported harvest against the more objective full-plot crop cuts. We replicate the main specification for the effect of recall length on farmer-reported harvest in the sample of plots for which a full-plot crop cut is available. Then, we quantify the reporting error (farmer-reported harvest quantity minus full plot crop cut quantity) and assess the effect of the recall length on the reporting error. We find a significant effect of the recall length on self-reported harvest as well as on reporting error (Fig. 6). This suggests that farmers are indeed over-reporting harvest with longer recall periods, at least in this sample. The effect of the survey timing on the objective estimate (the crop-cut, also reported in Fig. 6 for comparison) in contrast is not significantly different from zero.

[14] This does not mean that the estimate is correct, but rather suggests that it does not correlate with fieldwork timing (Carletto et al., 2011; De Groote and Traoré, 2005; Dillon et al., 2019; Goldstein and Udry, 1999; Keita and Carfagna, 2009; Kilic et al., 2017; Schøning, 2005).

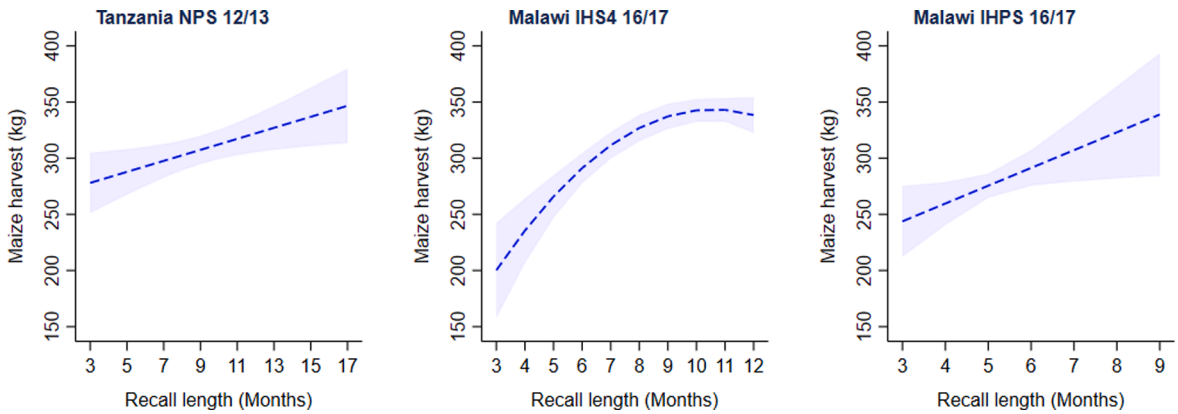

Fig. 4. Maize harvest quantity against recall length.

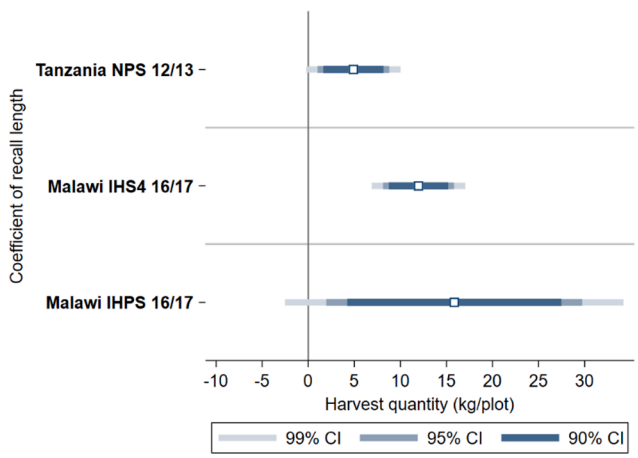

Fig. 5. Effect of recall length on reported maize harvest, point estimates.

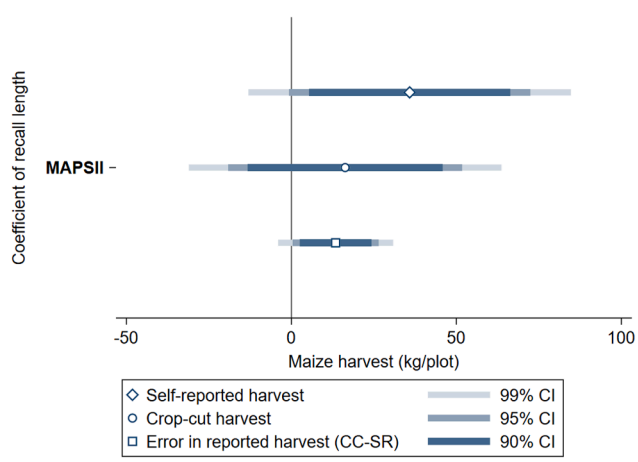

Fig. 6. Self-reported harvest and error in self-reported harvest (self-report – crop cut), Uganda MAPSII.

4.3. Labor

The literature reviewed in Section 2 reports inconsistent results with regard to the relationship between labor input estimates and the data collection method. We find that reported farm labor per plot increases significantly with recall length. The effect size is similar in Tanzania NPS 2012/13 and Malawi IHS4 at 1.6 (2.3 percent) and 1.2 person-days (2.8 percent) per plot per month of recall length, respectively (Fig. 7, Fig. 8, Table A.15, Table A.16, Table A.17).[15]

The concern with total labor days is that it is made up from different work activities with different timings. We therefore assess the recall effect separately for each type of activity: Land preparation and planting; weeding, ridging, and fertilizing; and harvest and post-harvest activities. Generally, the positive association between recall length and labor input holds across activities in all three datasets, though not all point estimates are significant (Fig. 7). Recall length has the largest and most significant level effect on reporting of labor dedicated to 'weeding, ridging, and fertilizing' (Fig. 7, Table A.18, Table A.19, Table A.20). This category is arguably the least salient and captures labor applied on the plot between planting and harvest, a period of several months, which make these activities particularly hard to recall correctly.

We also test family and hired labor input per plot separately. The total labor input results are driven by family labor. On the one hand, the number of family labor days make up a large share of total labor days. On the other hand, recall length has a significant effect on family labor days in all three datasets, but not on hired labor. Further, there is no significant effect of recall length on the number of family members reported to have worked per plot and per household (Fig. 9, Table A.15, Table A.16).

In recent studies, Arthi et al. (2018) and Gaddis et al. (2020, forthcoming) find two countervailing types of reporting bias on labor inputs. First, end-of-season recall leads to an over-estimation of reported labor inputs per plots ('recall bias'). Second, end-of-season recall leads to an under-counting of plots listed and household members reported to have worked on them ('listing bias'). Our results partially echo those findings: we find longer recall periods are associated with fewer plots listed and higher labor inputs reported. While we cannot benchmark our results against a more objective measure of farm labor, it is plausible that respondents here too are over-reporting as the recall period increases. A key difference is that we find no evidence of farmers reporting fewer members having worked on each plot or on the entire farm. Moreover, Arthi et al. (2018) and Gaddis et al. (2020, forthcoming) find a significant educational gradient in recall bias, while we find no evidence of education interacting with the recall effect.

4.4. Inputs

There is very little previous evidence to which to compare our results

[15] The comparison with Malawi IHPS 2016 panel is omitted because in that survey labor inputs are collected partially in the first visit (labor used for land preparation and planting, ridging and fertilizing) and partially in the second visit (harvest and post-harvest labor). Thus, there is no one unique recall period against which to evaluate total labor input, rendering the comparison meaningless.

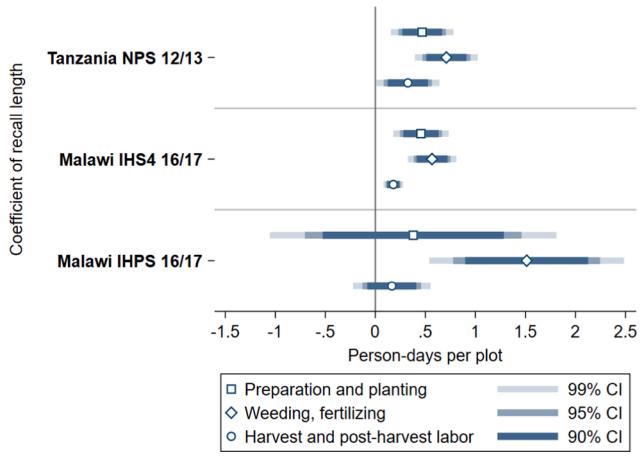

Fig. 7. Effect of recall length on labor input by activity, point estimates.

fertilizer quantities the longer the recall period.

Based on a Probit model, we find a small negative recall effect on the binary variable whether any inorganic fertilizer was applied in the case of Malawi IHS4 and IHPS 2016, and whether any organic fertilizer was applied in the case of Tanzania NPS 2012/13 (Fig. 10); that is, respondents are slightly less likely to report having used fertilizer as the recall length increases.

We use a linear OLS specification to estimate the effect of recall length on quantity of inorganic fertilizer applied per plot, conditional on fertilizer use.[16] The effect size is large in Malawi IHPS 2016/17 at 4.3 kg per plot per month (8.8 percent), and smaller in Malawi IHS4 at 0.9 kg (1.9 percent). In Tanzania NPS 2012/13, the coefficient has the same sign but is not statistically significant (Fig. 11, Table A.21, Table A.22, Table A.23). Low fertilizer application rates lead to a small sample of households using inorganic fertilizer on their land at 675 of 7,322 plots, which may explain the recall effect being insignificant. The effect of the recall length, in months, on the conditional use of inorganic fertilizer is depicted in Fig. 12.

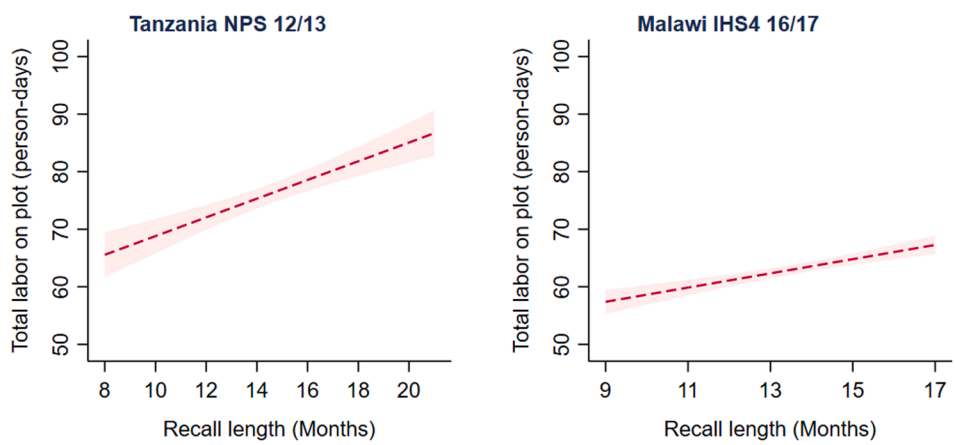

Fig. 8. Total labor on plot against recall length. Notes: Malawi IHPS 2016 omitted because labor input is collected in two visits so that the recall length variable cannot be constructed.

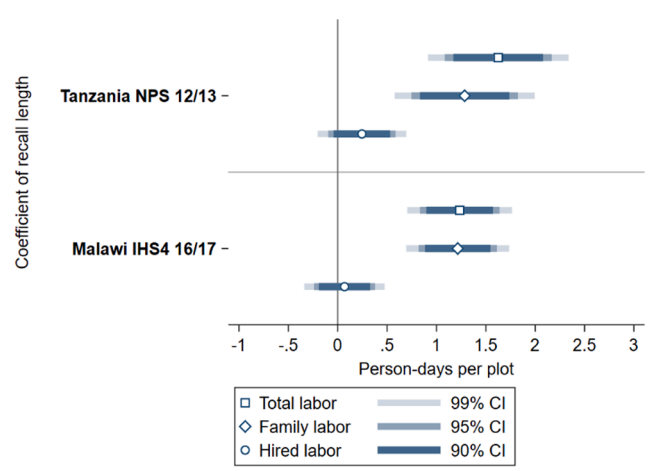

Fig. 9. Effect of recall length on labor inputs, point estimates, Notes: Malawi IHPS 2016 omitted because labor input is collected in two visits so that the recall length variable cannot be constructed.

on agricultural inputs. There are two main results regarding the effect of recall length on agricultural inputs (we analyze organic and inorganic fertilizer use in this study): first, as the recall length increases respondents are slightly less likely to report incidence of fertilizer use, that is whether or not any fertilizer was applied to a given plot. Second, when reporting any use of fertilizer, respondents tend to report higher

[16] We also run a Tobit on unconditional fertilizer quantity, finding no significant effect (Table A.21, Table A.22, Table A.23). This finding is unsurprising given the opposing signs of the results for conditional fertilizer quantity and for the binary of using fertilizer.

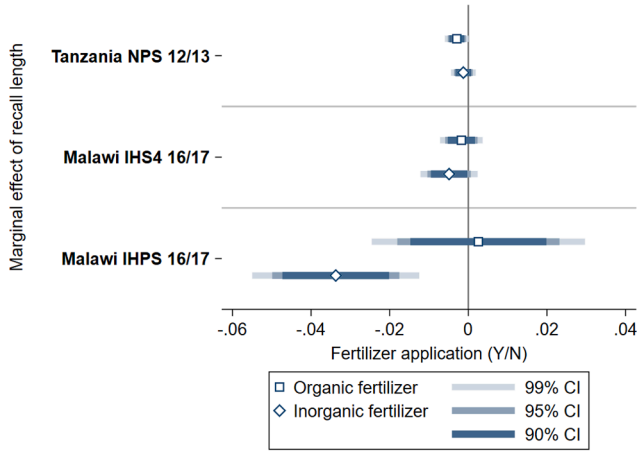

Fig. 10. Effect of recall length on whether respondent reported using fertilizer on plot, marginal effects.

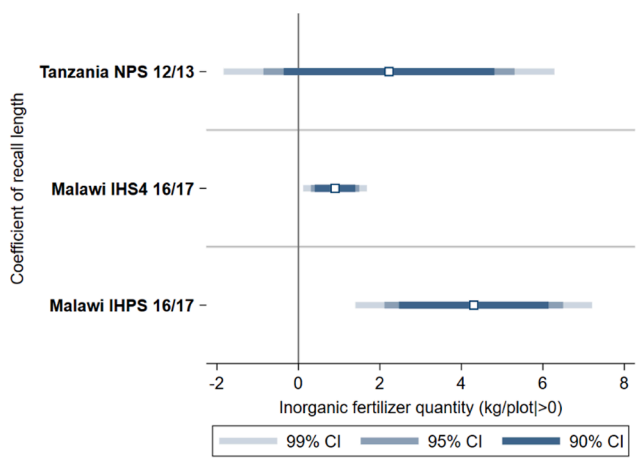

Fig. 11. Effect of recall length on reported fertilizer quantity per plot, point estimates.

In sum, the three continuous plot-level outcomes we have analyzed in this and the previous two sections – maize harvest, labor inputs, and quantity of inorganic fertilizer – are all positively correlated with recall length with similar effect sizes. This suggests that the same underlying error-generating process may be at play, making the presence of correlated non-classical measurement error a likely common phenomenon. Abay et al. (2019) discuss how the reporting error in plot area and harvest quantity can be correlated, posing specific problems for inference. Our analysis indicates that those problems may extend to other policy relevant analytical questions based on data generated through similar processes.

4.5. Robustness checks

The validity of the results presented above rests on the assumption that it is the length of the recall period what affects reporting, as opposed to other underlying factors that explain increased harvest, labor and other inputs, and fewer plots over time. We run several additional specifications as robustness checks to support that hypothesis, which are shown in appendix Tables A.25 to A.35.

A first set of robustness checks concerns maize quantity being reported in different states of the crop: at the time of harvest, maize is still on the cob. Before being sold, stored, or consumed maize grains may be removed from the maize cob and dried. In this process, the weight of the same harvest will vary depending on the state in which its weight is reported. Farmers will be more likely to report the harvest weight of maize on the cob immediately after the harvest and the weight of dried grains as more time passes between the harvest and the interview. This, in turn, could mean that the observed recall effect is the result of changes in reported harvest state over time. We assess this possibility using information on the harvest state – shelled (grain) and unshelled (on the cob) – which is available for Malawi IHS4 and IHPS 2016/17 data. To test whether this affects our results, we convert the quantity of unshelled maize (on the cob) to grain-equivalent shelled weight and then repeat the analysis with this new variable. The unshelled to shelled conversion factor was obtained from the MAPS experiment in which maize was weighed twice, before and after shelling. The results are robust to this adjustment for crop state (Table A.25, Table A.16, Table A.27).

Another set of checks concerns enumerators' characteristics and behavior. The results are robust to the exclusion of enumerator fixed effects, which we take to suggest that enumerator characteristics and behavior is not what is primarily driving our findings. Similarly, our results are generally robust to the exclusion of regional fixed effects as well as to the inclusion of the exogenous climatological and geospatial variables discussed in Section 3.3.

Another potential confounding effect is the interaction of interview timing and harvest timing: if households interviewed earlier had not harvested as much as households interviewed later simply because they had not finished harvesting, then this could offer an alternative explanation for our findings. While we drop households that have not finished their harvest from our sample to avoid the most severe confounding threat, this could mean that households with shorter harvest times and thus possibly smaller harvests select into the earlier months of fieldwork. To address this, we test a specification with the length of the harvest period (from beginning to end of harvest period) included but find this does not affect our results.

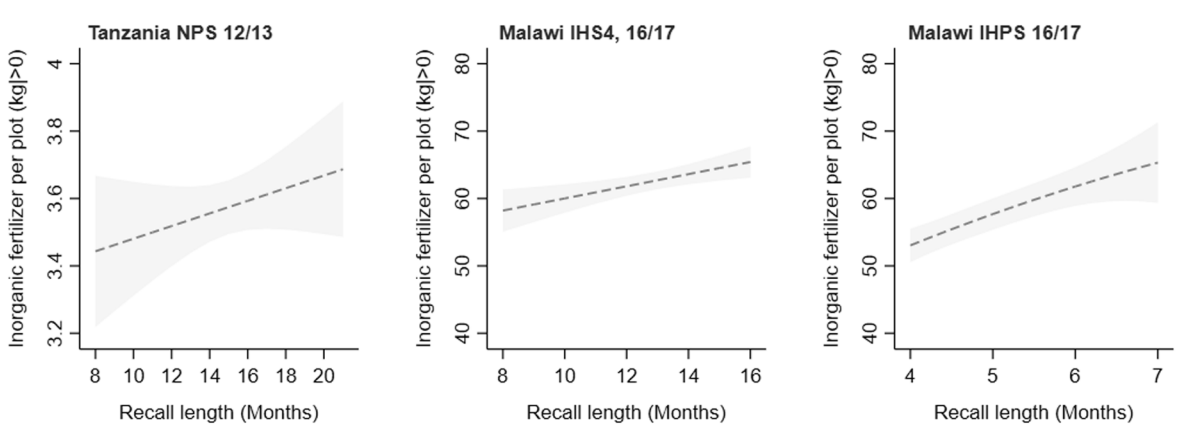

Fig. 12. Reported inorganic fertilizer amount (kg| > 0) against recall length.

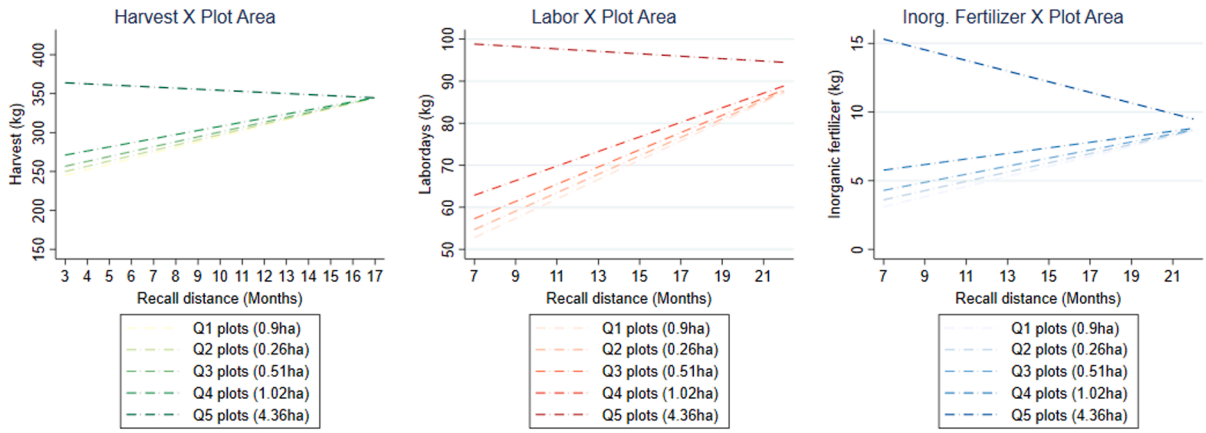

Fig. 13. Selected outcomes against recall length at five quintiles of plot area, Tanzania NPS 2012/13.

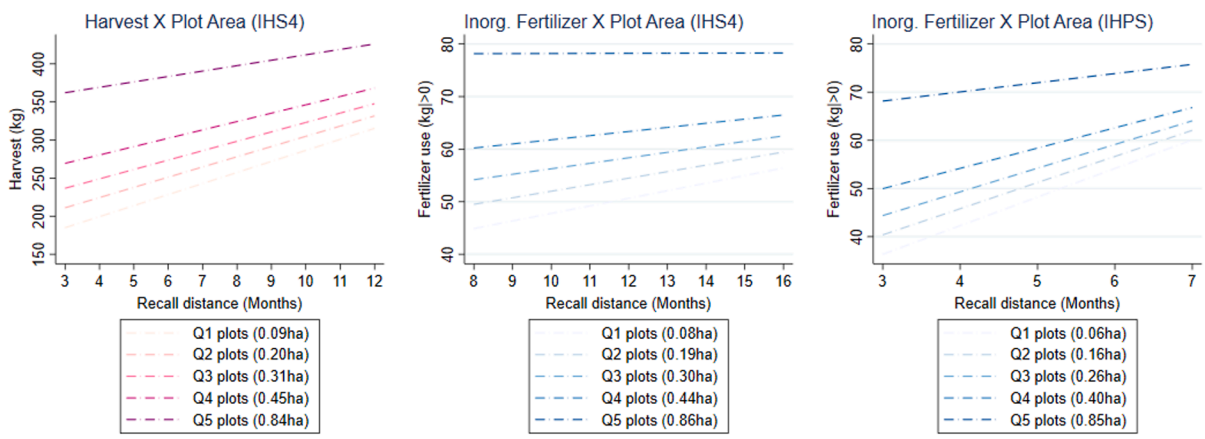

Fig. 14. Selected outcomes against recall length at five quintiles of plot area, Malawi IHS4 and IHPS 201.

Finally, split-off households and movers in the panel sample may deviate from the random interview timing and create a dynamic that confounds the results. We therefore drop split-off households and movers in an additional check, finding our results remain robust to this change in the sample.

4.6. Recall error and other characteristics

While our focus is on the effect of the recall length on data quality, we always control for respondent characteristics, including via interactions with the length of the recall. Most of the respondent characteristics (age, education, gender, whether respondents are also plot managers) have no significant impact on the direction or magnitude of the effect the recall length.[17] This also goes for respondent education which other studies had found to be correlated with reporting errors (e.g. Arthi et al., 2018).

Unlike respondent characteristics, plot size matters for the effect of recall length in a consistent manner. The recall effect on maize harvest quantity, labor, and fertilizer inputs is attenuated, and sometimes even reversed, on larger plots in Tanzania NPS 2012/13 (Fig. 13). Similarly, there is a significant interaction effect of recall length and plot size for harvest and fertilizer input in Malawi IHS4 and IHPS (Fig. 14).

4.7. Recall error and productivity measurement

This section explores the implications of the recall effect for agricultural productivity measurement, in which context the input and output variables discussed in this analysis are frequently studied. Interest is in three of the most common measures of agricultural productivity, yield (output per unit of land), output per day of work, and output per worker (Table 3). We have already estimated the recall effect for their components land, output, labor days and number of laborers. It is not clear ex ante what the recall effect on the composite variables will be, especially when the individual effects have the same sign. For instance, in output per day of work, both the numerator (output) and the denominator (labor-days) are positively associated with the recall length. The direction of the composite effect depends on which of the two dominates, and it is possible that there is no effect at all. (Table 3). As per the results of Abay et al. (2019) recalled earlier, in the cases where different measurement error in variables is correlated, correcting for it partially may in fact aggravate the bias.

Akin to the analysis discussed in previous paragraphs, we use as a proxy for recall length the time in months between the interview day and the first day of fieldwork. This proxy has two advantages in the case at hand. First, productivity measures are composite indicators whose components – output, land area, labor inputs – have each different timing patterns, so that there is no obvious single recall length measure. Second, time between the interview and the beginning of fieldwork has a simple practical interpretation and can be determined by survey designers. We test the effect of the recall length both at the plot level and

[17] Plots listed in Malawi IHS4 cross section is the one case in which there is a significant interaction effect between respondent characteristics and recall length. Female respondents list fewer plots than male respondents (−0.52*** plots), but the effect of recall length is smaller among female respondents (−0.05** plots/month as opposed to −0.076*** plots/month). Female respondents also report less harvest and input quantities relative to male respondents.

Table 3
Selected productivity measures.

Productivity measure	Construction	Estimated individual recall effects	Estimated recall effect on composite variable
Yield (Y/T)	Production per plot/ farm (Y) / Land area per plot/farm (T)	Y↑, T∅	Positive
Output per day worked (Y/L)	Production per plot (Y) / Total labor days per plot/farm (L)	Y↑, L↑	Positive
Output per worker (Y/I)	Production per plot (Y) / household members per plot/farm (I)	Y↑, I∅	Positive

aggregated at the farm level in the Malawi IHS4 and Tanzania NPS 2012/13 data using an OLS specification as in equation (5).[18]

We find that recall length is positively associated with the productivity measures of interest, both at the plot and at the farm level (Fig. 15; Table A.36; Table A.37). Maize yield at the plot-level increases by 3.1 percent with each additional month of fieldwork in Tanzania NPS 2012/13 and by 5.1 percent in Malawi IHS4 (at the farm-level: 2.2 percent and 5.2 percent, respectively). Output (maize harvest) per day of labor input is also generally positively associated with fieldwork length, with the effect ranging from null to 3.1 percent per month (Tanzania NPS 2012/13: null effect at the plot-level, 1.9 percent at the farm-level; Malawi IHS4: 2.7 percent at the plot-level, 3.1 percent at the farm-level). Finally, output per worker is also positively associated with fieldwork length, with the effect size ranging between 2.4 and 5.1 percent per month (Tanzania NPS 2012/13: 2.4 at the plot-level, 2.5 percent at the farm-level; Malawi IHS4: 5.1 percent at the plot-level, 3.0 percent at the farm-level).

5. Discussion and policy implications

In this paper, we set out to evaluate the impact of the recall length and related design choices on the quality of key agricultural input and output data – land, labor, fertilizer, and production. Our results demonstrate, consistently across the three datasets used in the analysis, that the survey-based estimates of these variables depend on the length of the recall period, which indicates the presence of nonrandom measurement error of economically significant size. With longer recall periods, farmers report higher quantities of harvest, labor and fertilizer inputs, all three of which are recorded at the plot-level in our datasets. At the same time, farmers list fewer plots as the recall period increases. We have argued that it is plausible that farmers over-estimate plot-level outcomes – harvest, labor and fertilizer inputs – while it is also plausible that they forget some of their more marginal plots, as their memory decays due to longer recall periods. We also show that the recall length has a meaningful impact on agricultural productivity measurement.

The reliability of agricultural data matters for policy effectiveness. Policymakers and the international community recognize agricultural input, output, and productivity outcomes as integral to agricultural growth and food security, and a large body of empirical evidence supports this view. These outcomes have therefore been made priorities within several development frameworks. Under the Sustainable Development Goals (SDGs), target 2.3 is to double the agricultural productivity and incomes of small-scale food producers by 2030. The target is monitored through indicator 2.3.1, volume of production per labor unit – that is, one of the productivity measures we found were subject to measurement error related to the recall length. Similarly, African Union's (AU) Comprehensive Africa Agriculture Development Programme (CAADP) was conceived in 2003 to achieve economic growth, eliminate hunger, and reduce poverty through concerted investments in African agriculture. CAADP adopted the Malabo Declaration in 2014 in which AU member states resolved, among other things, to double agricultural productivity and pledged to allocate at least 10 percent of public expenditure towards agriculture. The CAADP results framework monitors progress towards its goals through a set of priority indicators and guides member countries' strategic investment decisions. CAADP target 2.1 – Increased agriculture production and productivity[19] is comprised of five priority indicators, focusing on growth in agricultural production volume and value added as well as land and labor productivity. For instance, priority indicator 2.1.5 measures yield for the five AU priority commodities, of which maize is one; indicator 2.1.2 tracks the level of agricultural production.

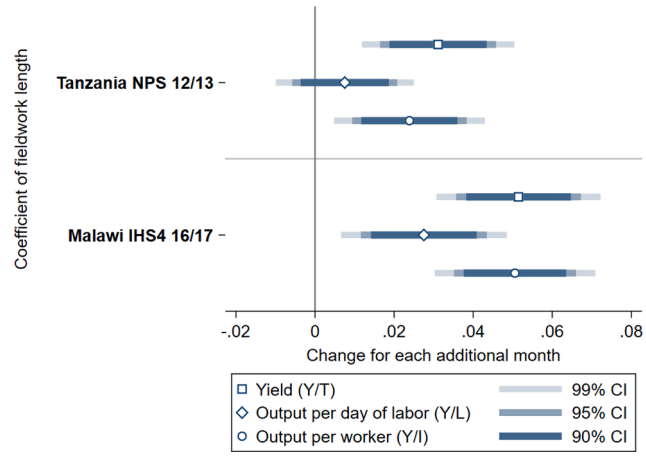

Fig. 15. Impact on key productivity measures, plot-level, Notes: Malawi IHPS 2016 omitted because output, labor input, and land input information is collected in two separate visits so that a single recall or fieldwork length variable cannot be constructed.

The results of our analysis show that recall error can lead to unreliable measurement of all these indicators at an economically significant level. This is especially true when it comes to monitoring their evolution over time. For illustration, maize yields in Africa (reflected in CAADP 2.1.5) have grown at an average annual rate of 0.8 percent between 2000 and 2017, according to FAOSTAT. Maize production (reflected partially in CAADP 2.1.2) has grown by an average of 3.9 percent per year during the same period. In comparison, our analysis indicates that an additional month in recall length can lead to a change of between 2 and 5 percent in reported maize yields and 2 and 7 percent in the reported maize harvest quantity. Further, an annual growth rate of around 7 percent would be required to achieve SDG target 3.2 of doubling labor productivity within about 10 years. We find an additional month of recall length is associated with a 2 to 3 percent increase in reported output per unit of labor. What this shows is that the recall effect is likely to introduce an amount of variability that makes tracking progress towards achieving SDG targets and CAADP goals challenging and sensitive to the survey design choices that affect the length of recall. Policy and investment priorities based on these indicators risk being ineffective or misguided. Improving data quality is critical to minimize this risk.

The results lend support to ongoing efforts to mainstream more objective 'gold standard' measures of key variables into the design and implementation of surveys. The use of GPS devices to measure the area

[18] The Malawi IHPS 2016/17 data is omitted because the two-visit structure of that survey makes time between interview and beginning of fieldwork difficult to interpret.

[19] CAADP Indicator 2.1. *Increased agriculture production and productivity* contains the following sub-indicators: 2.1.1 *Agriculture value added*; 2.1.2 *Agriculture production index*; 2.1.3 *Agriculture value added per agricultural worker*; 2.1.4 *Agriculture value added per hectare of arable land*; 2.1.5 *Yields for the five AU priority commodities* (cassava, yams, maize, meat, and cow milk).

of agricultural plots is already considered best practice and incorporated into many surveys concerned with agricultural productivity and land use. In the field of production measurement, the practice of crop cutting is used in many national production estimations and other surveys, and our findings offer strong reasons for doing so systematically. Finally, when it comes to accurately measuring labor and fertilizer input, as well as counting all plots used in agricultural production, the use of diaries appears worth exploring. With cost considerations in mind, these more objective measures may be taken on a subsample of farms or plots and then used to benchmark and correct the self-reported measures in the full sample, in which process recall length would be a variable to consider.

One strategy to improve agricultural data reliability by shortening recall periods and recall length variation in household and farm surveys is to field two visits, one in the post-planting and one in the post-harvest period, rather than just one. While this strategy has cost implications, as field teams need to visit farms twice rather just once, it has the additional advantage of providing countries with more frequent and timely data on planting in the ongoing agricultural season. Another avenue to explore is the use of higher-frequency phone surveys to collect data on input and even output variables, though the data quality implications of such a survey design ought to be rigorously tested before mainstreaming. Finally, survey timing could be randomized, to the extent that this is in line with other survey objectives. Results from farms or plots visited earlier in the process could then be used in a similar way as the 'gold standard' measures discussed above to benchmark and correct results based on longer recall periods.

In sum, there is a need for survey practitioners and agencies charged with collecting agricultural sector data to carefully consider the data quality implications of survey design choices that implicitly affect the length of the recall period for the respondent, and to develop and promote the adoption of survey designs that can ensure data quality without excessive implications for the survey budgets and logistics.

CRediT authorship contribution statement

Philip Wollburg: Conceptualization, Methodology, Formal analysis, Writing - original draft, Visualization, Writing - review & editing. **Marco Tiberti:** Conceptualization, Methodology, Writing - original draft, Writing - review & editing. **Alberto Zezza:** Conceptualization, Methodology, Writing - original draft, Writing - review & editing.

Appendix A. Supplementary material

Supplementary data to this article can be found online at https://doi.org/10.1016/j.foodpol.2020.102003.

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